What Are the Opportunities and Challenges of Using AI in Medical Education in Vietnam?

Abstract

Artificial intelligence (AI) has the potential to transform medical training through adaptive learning, immersive simulations, automated assessments, and data-driven insights, offering solutions to persistent issues such as high student-to-faculty ratios, overcrowded classrooms, and limited clinical exposure. Globally, many universities have already embedded AI literacy and competencies into undergraduate, postgraduate, and continuing education programs, while in Vietnam, the use of AI in medical education remains limited and fragmented. Most students have little formal exposure to AI, and empirical evidence on faculty or institutional readiness is scarce. Experiences from other countries, including Malaysia, Palestine, and Oman, demonstrate that incremental adoption and faculty development can facilitate cultural acceptance and curricular innovation, providing useful lessons for Vietnam. At the same time, significant barriers remain. These include inadequate infrastructure in provincial universities, low levels of AI literacy among both students and educators, underdeveloped regulatory and ethical frameworks, and resistance to pedagogical change. Cost-effectiveness and sustainability are additional concerns in a middle-income context, where upfront investments must be balanced against long-term benefits and equitable access. Advancing AI in Vietnamese medical education will

therefore require a coordinated national strategy that prioritizes infrastructure, AI literacy, faculty development, quality assurance, and sustainable funding models, alongside ethical and legal safeguards. By addressing these key foundations, Vietnam can harness AI not only to modernize medical education but also to strengthen preparedness for a digitally enabled health workforce.